

I Got It from My Mama:

Maternal Immune Activation and Reward Learning

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Abstract

Sign- and goal-tracking are distinct behavioural phenotypes characterised by the direction of attention after presentation of a condition stimulus signalling an oncoming reward. Sign-tracking is associated with several maladaptive behaviours including increases in risk taking, novelty seeking, impulsivity and drug-related behaviours, as well as cognitive impairments such as deficits in attention. The underlying neural circuitry implicated in sign-tracking is also implicated in neurodevelopmental disorders such as ASD and schizophrenia, which share maternal immune activation as a common risk factor. The present study aimed to explore the effect of polyIC-induced maternal immune activation on sign-tracking behaviour in adolescent rats. Pregnant Sprague Dawley rats were injected with either viral mimic polyIC or saline on GD12. Offspring were subjected to a Pavlovian conditioning paradigm for 5 consecutive days from PND45. Sign- and goal- tracking responses were measured on the fifth day. It was hypothesised that offspring of polyIC-treated mothers would show greater sign tracking behaviour than the offspring of saline-treated mothers, but no significant group effect was found.

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Classical conditioning is the learning process whereby the repeated pairings of a neutral stimulus (NS) with an unconditioned stimulus (US) causes an animal to respond to the NS as if it were the US, in the absence of the original US. Pavlov (1927) first observed this phenomenon where the presentation of food (US) would cause dogs to salivate (unconditioned response, UR), and repeated pairings of the US with a bell sound (NS) would cause dogs to salivate at the sound of a bell (conditioned response, CR) in the absence of food (US) (Pavlov 1927). This has become a fundamental phenomenon in behaviourism and is often referred to as the Pavlovian Conditioned Approach (PCA) paradigm (Colaizzi et al., 2020; Fitzpatrick & Morrow, 2016). Rodent studies into Pavlovian conditioning have highlighted the emergence of two distinct phenotypes in response to a conditioned stimulus. When rats are exposed to the PCA paradigm involving a lever (CS) predicting the delivery of a food reward a short time later (US), the predictive value of the CS produced two distinct conditioned responses. Following the presentation of the lever, a subpopulation of rats consistently direct their attention to the location of the food delivery. This subpopulation of rats is termed “Goal-Trackers” (GTs), as their conditioned response is directed towards the reward or ‘goal’. A second subpopulation of rats tend their attention to the actual lever, the CS, earning the title of “Sign-Trackers” (STs), because they direct their attention to the stimulus that signals the reward, rather than the location of food delivery (Colaizzi et al., 2020). It is thought that for STs, the stimulus not only attained predictive value, but was also attributed incentive salience (Fitzpatrick & Morrow, 2016; Meyer et al., 2012). This attribution of incentive salience has three defining characteristics. Firstly, the stimuli bias attention towards them, meaning they become attractive, and the subject approaches them. Secondly, the stimuli themselves become desirable objects, motivating persistent

reward-seeking behaviour. thirdly, incentive stimuli are capable of generating a motivational state that triggers these behaviours (Meyer et al, 2012).

The emergence of these two distinct phenotypes is not considered a stand-alone difference in response to rewarding stimuli, but rather a characteristic of a larger collection of behavioural differences. Relative to GT rats, ST rats display increases in novelty seeking, risk taking and impulsivity, show a higher propensity for substance abuse, and display deficits in sustained attention (Colaizzi et al., 2020). Stead and colleagues (2006) investigated selectively breeding Sprague-Dawley rats based on their responsiveness to a novel environment. When tested in a Pavlovian reward paradigm, high responders to novelty (bHR) consistently displayed ST behaviour, while low responders (bLR) consistently displayed GT behaviour. bHR also display higher rates of novelty seeking and drug abuse, and self-administer cocaine and amphetamine at higher doses than bLR. (Stead et al., 2006). The ability to selectively breed for these phenotypes and the robustness of the associated characteristics is evidence to support a genetic component to the ST/GT phenomenon. ST rats have also been found to be more impulsive. Following PCA training, Lovic and colleagues (2011) subjected ST and GT rats to three tests of impulsivity, two of which measured impulsive action (2- choice serial reaction time task & differential reinforcement of low rates of responding task) and one of impulsive choice (Delay Discounting choice procedure). STs showed worse performance (were more impulsive) than GTs on the two tasks of impulsive action, but were less impulsive on the Delay Discounting task, a measure of impulsive choice. (Lovic et al., 2011). ST rats also show poorer performance on a sustained attention task than GT rats. (Paolone et al., 2013).

The neural mechanisms underlying ST/GT differences are thought to be associated with the mesolimbic dopamine system, which is the primary neural network involved in reward approach, receipt and learning (Schultz, 2015). These processes are mediated by dopaminergic projections between cortical structures such as the orbital frontal cortex (OFC) prefrontal cortex (PFC) and anterior cingulate cortex (ACC), and subcortical structures such as the ventral tegmental area (VTA) and nucleus accumbens (NAc) of the (ventral) striatum (Greene et al., 2019). The receipt of a reward triggers an initial dopamine release within this mesolimbic dopamine system which, following Pavlovian conditioning, shifts to a predictive cue of reward (Flagel et al. 2011). To better determine the role of dopamine in the attribution of incentive salience observed in STs, Flagel and colleagues (2011) used fast-scan cyclic voltammetry (FSCV) to measure patterns of phasic dopamine signalling within the NAc. The CS acted as an equivalent predictor of reward in both STs and GTs, as demonstrated by similar changes in latency for their respective CRs during learning. CS-evoked dopamine release increased in sign-tracking rats relative to controls, with no such effect observed in goal-trackers. Furthermore, US-evoked dopamine release in sign-trackers was significantly lower than goal-trackers following acquisition. This provides evidence to suggest fundamental differences in the pattern of dopamine release to reward-related stimuli between the two groups (Flagel et al., 2011).

Neurobiological studies (Flagel et al., 2011; Kuhn et al., 2018; Paolone et al., 2013) have demonstrated that the distinction between sign tracking and goal tracking behaviour represents an imbalance between top-down and bottom-up processes. Goal-tracking behaviour is thought to be mediated by top-down processes, whereas sign-tracking behaviour is mediated by bottom-up, subcortical processes, with a greater reliance on the “motive circuit” which is responsible for integrating information

regarding a motivationally salient event. This circuit comprises the exchange of information between subcortical structures mediating emotional components (fear, reward) and the PFC, which mediates goal-directed behaviour by exerting inhibitory control to attend to the most meaningful stimuli (Kuhn et al., 2018). Only when a reward cue is attributed with incentive salience, such as in STs, do we see activation of this circuit (Flagel et al., 2011). Within the ST/GT model of addiction, it is dysregulation of this circuit that contributes to addiction and relapse (Flagel et al., 2011; Kalivas & Volkow, 2005). Pitchers and colleagues (2017) demonstrated that STs, as determined by their responding to a cocaine cue, exhibited increases in dopamine levels within the PFC when presented with that same cue. No such changes were observed in acetylcholine (ACh) levels. Conversely, GTs exhibited ACh increases in the PFC when presented with the same cue and displayed no changes in dopamine levels (Pitcher et al., 2017). These findings provide further evidence that the dopamine-dependent encoding of incentive motivational value is separate from the dopamine-independent encoding of the value of reward cues (Kuhn et al., 2018).

This mesolimbic pathway has been implicated in several disorders, including autism spectrum disorder (ASD) and Schizophrenia. ASD is a neurodevelopmental disorder characterised by deficits in social functioning and routine, repetitive behaviours. The social motivation hypothesis of ASD suggests that symptoms arise from a lack of attention to social stimuli due to a lack of social motivation attributed to decrease in the rewarding value of social stimuli (Scott-Van Zeeland et al., 2010). Schmitz and colleagues (2008) used event-related fMRI to examine neural responding to reward in ASD. They hypothesised that behavioural abnormalities in people with ASD result from a lack of perceived reward feedback and thus ASD individuals would show a different pattern of activation in frontostriatal and frontolimbic structures. A

continuous performance test with a monetary incentive was employed and neuroimaging data was collected during reward achievement. ASD individuals showed significantly greater brain activation in the left ACC than controls. This increased activation may be a compensatory mechanism to overcome the neurobiological deficits associated with the disorder. Thus, the authors concluded that activation differences in areas responsible for attention and arousal may mediate deficits in social behaviour (Schmitz et al., 2008). Dichter and colleagues (2012) was the first to investigate the effect of different incentive types on mesolimbic reward circuitry in individuals with ASD. An incentive go/no-go task was employed comparing monetary and ASD-salient object rewards. This task allows for assessment of reward anticipation, initiated by cue signals, and subsequent goal directed behaviour (Kohl, 2013). fMRI of blood-oxygen level-dependent (BOLD) revealed that the ASD group showed decreased NAc activation during monetary rewards but not objects (Dichter et al., 2012). Furthermore, a robust interaction effect was found between group and reward type on bilateral NAc during anticipation of rewards and ventral PFC during outcomes. Thus, the authors concluded that differential responding in these regions is dependent on reward type. Overall, the ASD group showed hypoactivation of the reward circuitry during money incentives but not objects, revealing a double dissociation of reward type and temporal phase of reward (Dichter et al., 2011). Kohl (2013) replicated this study with monetary and social rewards to assess the social motivation hypothesis of ASD. Consistent with Dichter and colleagues, (2012) the ASD group showed overall hypoactivation in mesocorticolimbic circuitry to both reward types in ASD and lower NAc activation in response to monetary rewards (Kohl 2013). However, this study did not find greater malfunctions in reward circuitry in response to social rewards than monetary rewards in ASD group, in contrast to the findings of Scott-Van Zeeland.

These findings suggest that preferential activation of the NAc observed in healthy controls reflects a greater attribution of incentive salience to the reward (Berridge et al., 2009).

The studies outlined above suggest that functional mesolimbic impairments in ASD may extend to a range of rewards beyond just social stimuli. These reward processing impairments in ASD align with the social motivation theory to account for the social impairments characteristic of the disorder. Greene and colleagues (2019) proposed that, although heterogenous, a large portion of ASD cases may be a result of neuroinflammation disrupting the development of mesolimbic dopaminergic systems. In addition to the clinical evidence, there is an abundance of preclinical evidence to support the mechanistic role of neuroinflammation in the reward processing impairments in ASD. Specifically, there is a bidirectional relationship between neuroinflammation and dopamine signalling (Greene et al., 2019). Neuroinflammatory processes in ASD have been widely studied in the context of maternal immune activation (MIA). Epidemiological studies have demonstrated that the risk of a child developing ASD is significantly increased if mothers were hospitalised with an infection during pregnancy (Patterson, 2011; Atladóttir, 2010). Specifically, a viral infection in the first trimester or bacterial infection during the second trimester (Atladóttir, 2010). It is hypothesised that the antibodies that developing fetuses are exposed to in utero elicit neurotoxic effects of fetal brain tissue that is treated as a foreign pathogen (Greene et al., 2019). Maternal infection can be modelled in rodents through administration of a bacterial mimic such as lipopolysaccharide (LPS) or viral mimic such as poly-inosinic:polycytidylic acid (polyIC) during gestation. This results in significant dysregulation of the striatum and PFC and are well-established as preclinical rodent models of inflammation in ASD (Greene et al., 2019).

Another disorder that is particularly sensitive to maternal infection is schizophrenia, a brain disorder characterised by both positive and negative symptoms. Positive symptoms include hallucinations, delusions and catatonia, whereas negative symptoms include social withdrawal, apathy, anhedonia and poverty of speech (Kneusel et al., 2014). Similar to ASD, schizophrenia has strong epidemiological associations with maternal infection and an increase in expression of cytokines in the brain. Both disorders share genetic risk factors in relation to immune function and it is thought that MIA might potentiate the effect of these genes (Kneusel et al., 2014). Establishment of the polyIC model allowed for the discovery that MIA was responsible for the observed changes in offspring and not the virus itself (Smith et al., 2007). The gestational timing of polyIC exposure has been shown to result in different behavioural dysfunction, with mid-gestation administration resulting in prepulse and latent inhibition (event-related potentials specific to schizophrenia) and late-gestation administration resulting in more pronounced impairments in cognitive flexibility. Other behavioural abnormalities such as selective attention, exploratory behaviour, and working memory were not specific to time of administration (Kneusel et al., 2014).

Schizophrenia has one of the largest prevalence of comorbid substance abuse of any disorder, with nearly 50% of patients affected by substance abuse (Menne & Chesworth, 2020). It is widely understood that the neuropathology of schizophrenia involves disturbances in the reward circuitry involved in positive reinforcement, incentive motivation, behavioural inhibition and addiction. Blocking of dopamine receptors in patients with schizophrenia exerts antipsychotic effects, whereas NAc dopamine elevation through psychostimulants results in psychotic symptoms in both schizophrenic patients and healthy controls (Chambers et al., 2001). The primary addiction hypothesis posits that the tendency to abuse drugs is not a result of

schizophrenia symptomology, but rather neurobiological disturbances in incentive motivational processes which also underly other behavioural disturbances observed in schizophrenia (Chambers et al., 2001).

It is evident throughout the literature that the neurobiological mechanisms underlying differences in sign- and goal- tracking behaviour are implicated in disorders such as ASD and schizophrenia. The risk of both of these disorders is significantly increased following MIA which can be modelled in rodents with a bacterial (LPS) or viral (polyIC) mimic. An increase in inflammatory markers from MIA may affect neurodevelopment of the mesolimbic dopamine circuit, resulting in dysregulation between cortical- and subcortical structures involved in the attribution of incentive salience. Behavioural tests of reward processing in ASD samples suggest an impairment in attributing incentive salience to rewards, which, in a Pavlovian conditioning paradigm, may present as a reduction in goal tracking behaviour. Furthermore, the behavioural impairments and underlying neural substrates in schizophrenia bear several similarities to the behavioural profile of ST rats. The present study aimed to explore the potential effect of polyIC-induced maternal immune activation on sign-tracking behaviour in adolescent rats. It was hypothesised that offspring of polyIC-treated mothers would show greater sign tracking behaviour than the offspring of saline-treated mothers.

Method

Subjects

Eight adult female and eight adult male Sprague-Dawley rats were paired in home cages with grated flooring to assess for vaginal plugs. Cages were checked every day for vaginal plugs to mark gestation day (GD) 1. Animals were then separated and

housed individually. On GD12, dams were randomly assigned to polyIC (n=4) or saline (n=4). Temperature and weight were measured immediately before subcutaneous injection of either 5mg/kg polyIC or 1mL/kg of saline. Weight was measured again 24 hours later, and temperature measured two, four and 24 hours after injection to ensure the polyIC had taken effect, represented as an increase in body weight and slight increase in body temperature. Two males and two females from each litter served as subjects (n=32). Rats were housed in groups of two, based on sex and treatment, in individually ventilated polycarbonate cages. Home cages were lined with saw dust and contained a rectangular plastic tunnel and wooden chewing rod. Animals were housed on a 12-hour light/dark cycle with temperature set at 19-21 degrees Celsius and humidity set at 55%. Experimentation took place during the dark cycle. Food restriction began at post-natal day (PND) 42. Females were fed 13g/kg and males 16g/kg of chow per day, increasing by 1g every two days, to maintain weight at 85-90% of expected free-feeding weight. These Increases were necessary to account for expected growth of adolescents. Food was delivered to the cages once daily, varying in length of time following experimentation so the animals would not learn to predict feeding from experimentation.

Apparatus

The Pavlovian conditioning paradigm was administered using standard operant chambers (Med Associates Inc., St. Albans, VT, USA) measuring approximately 31cm x 31cm x 25cm. a sugar pellet dispenser was located at floor level on the left-most third of the front interface panel, serving as the goal (location of reward delivery). A standard 100mA white light was situated in the top right third of the front interface panel, serving as the cue. MED-PC® software was used to run the paradigm. White, odourless, 45mg sugar pellets (Bio-serv ® Flemington, NJ, USA) served as the reward.

Pavlovian Conditioning Paradigm

Pavlovian conditioning consisted of one-hour daily session for five consecutive days, starting on PD45. light (CS, sign) was presented for 2 min at random time intervals with an average intertrial interval of 2 – 3 minutes. Presentation of the light was accompanied by an immediate delivery of four sucrose pellets, followed by another four at one minute. There was a total of 14 CS+ trials per session.

Data Analysis

Video recordings of the fifth and final session, (PND49) for each rat were analysed. Goal-responding was operationalised as the number of nose pokes into the pellet delivery system (goal) during light-on trials (CS+). Sign responding ($n(\text{sign})$) was operationally defined as the number of contacts made to the light with the nose or paw during light-on trials (CS+). Results were analysed using jamovi (2022) computer software version 2.3. the alpha level of significance was set at $p < 0.05$.

Results

Sign tracking behaviour was measured as the proportion of sign responses during CS+ trials ($p(\text{sign})$). This was calculated by dividing the total number of sign responses by the total number of responses during CS+ trials ($p(\text{sign}) = n(\text{sign})/(n(\text{sign})+n(\text{goal}))$). The mean and standard deviation (SD) of $p(\text{sign})$ and total number of responses for each treatment group is presented in Table 1.

Table 1

Mean (M) and standard deviation (SD) of dependent variables (DV) $p(\text{sign})$ and total responses for each treatment group

DV	Saline				polyIC			
	Male		Female		Male		Female	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
$p(\text{sign})$	0.170	0.106	0.144	0.074	0.155	0.174	0.205	0.072
Total responses	137	117	112	42.5	186	74.8	95.4	28.0

Figure 1 shows the mean number of total responses during CS+ trials for each group, and the mean proportion of goal and sign responses. A two-way ANOVA was conducted to examine the effect of sex and treatment group on p(sign). There was no significant effect of treatment on p(sign) ($F = 0.3356, p = 0.567$). There was no significant effect of sex on p(sign) ($F = 0.0903, p = 0.766$). There was no significant interaction between sex and treatment group ($F = 0.8939, p = 0.353$). The full ANOVA table is available in Appendix A.

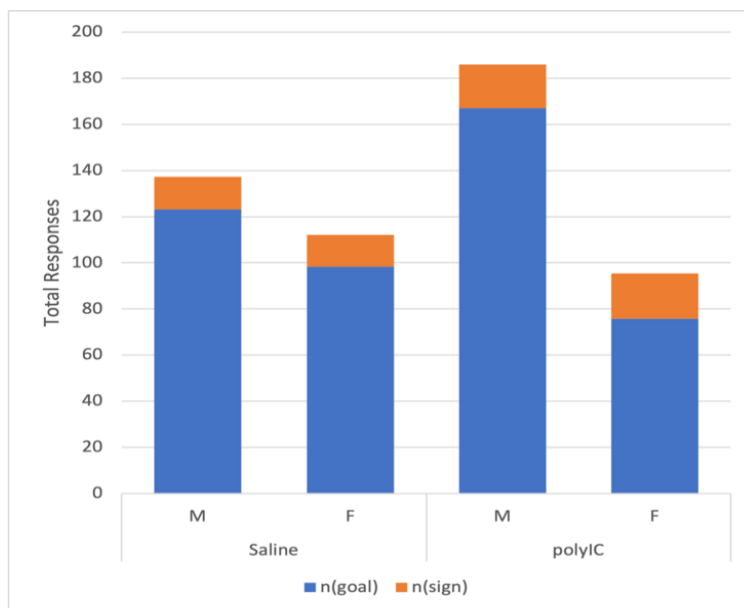


Figure 1. Mean number of goal ($n(goal)$) and sign ($n(sign)$) responses for males and females in each maternal treatment group. Area of the bars show these values as a proportion of mean total responses for each group (Y axis)

A two-way ANOVA was conducted to examine the effect of sex and treatment group on total number of responses. There was no significant effect of treatment on total number of responses ($F = 0.376, p = 0.545$). There was a significant effect of sex on total number of responses ($F = 4.919, p = 0.035$). There was no significant interaction between sex and treatment group ($F = 1.575, p = 0.220$). The full ANOVA table is available in Appendix A.

Discussion

The aim of the present study was to explore the effect of polyIC-induced maternal immune activation on sign-tracking behaviour in adolescent rats. It was hypothesised that the offspring of polyIC-treated mothers would show greater sign-tracking behaviour than the offspring of saline-treated mothers. There were no significant differences in the proportion of sign responses between polyIC and saline treatment groups, suggesting maternal immune activation has no effect on sign tracking behaviour. This finding unexpected given the common neural mechanisms underlying sign tracking behaviour and schizophrenia, a disorder frequently implicated in maternal immune activation. Although a maternal immune response was achieved by the polyIC, as evidenced by subsequent increases in body temperature and weight, it may be that the time of treatment administration (GD12) was outside of optimal window for the neuroinflammatory response to alter development of the mesolimbic system. Behavioural abnormalities most reflective of sign tracking behaviour, such as selective attention and exploratory behaviour, have not previously been associated with specific temporal windows of maternal infection (Kneusel et al., 2014). Despite this, maternal immune activation still may result in dysregulation of the mesolimbic reward system at a different point in gestation, not investigated by the present study. Greene and colleagues (2019) also provided evidence suggestive of a dissociation between the timing of immunological insult subsequent striatal dopamine, with earlier insults resulting in hyperactive striatal dopamine and later insults having the opposite effect. Future studies looking to investigate the effects of maternal infection on sign tracking behaviour may benefit from comparing different temporal points of polyIC administration.

In contrast to previous studies using a Pavlovian conditioning paradigm (Fitzpatrick & Morrow, 2016; Meyer et al., 2012; Paolone et al., 2013; Stead et al., 2006) which found varying degrees of sign- goal, and intermediate responding, the present study did not find a single subject who reliably displayed sign tracking behaviour. Only one subject responded intermediately between sign- and goal- tracking, the rest strongly displayed goal-tracking behaviour, regardless of treatment condition. There are several possible explanations for this finding. Firstly, previous studies primarily used adult rats, whereas the present study used adolescent rats, with their sign-tracking behaviour measured at PND49. Rode and colleagues (2020) demonstrated that despite displaying higher rates of behaviours associated with sign tracking (novelty exploration, impulsivity, substance abuse) adolescent rats actually display less sign-tracking behaviour and greater sign tracking behaviour than adults in a Pavlovian conditioning paradigm (Rode et al., 2020). Secondly, Colaizzi and colleagues (2020) described adult rats as displaying more sign tracking behaviour than adolescents of the same genetic strain. However, adolescent rats are more sensitive to food restriction and display increased sign tracking in response to stress (Colaizzi et al., 2020), so environmental factors may need to be more tightly controlled when testing adolescent rats in a Pavlovian conditioning paradigm than is required for adults. Despite appropriate food-restriction measures were taken to ensure the animals weighed 90% of their projected free feeding weight, food restriction in general (necessary for reward learning) could have increased sign responding in the present study, but this was not found. Finally, adolescent rats have lower basal dopamine levels in the NAc than adult rats and thus may not experience the attribution of incentive salience to the same extent as adults, minimising the potential for sign tracking behaviour (Colaizzi et al., 2020).

Another potential explanation for the high rate of goal tracking observed in the present study may lie within methodological differences between the present study and previous sign tracking literature. In previous studies (Fitzpatrick & Morrow, 2016; Meyer et al., 2012; Paolone et al., 2013; Stead et al., 2006), the reward cue (CS, light) was presented a short duration before presentation of the reward, and sign-/goal-responding was measured during CS presentation (before reward receipt). In contrast, the present study paired the reward with the cue simultaneously and measured responding at the same time. This may have led to an overestimation of goal-responding since retrieval of the reward was counted as a response to the cue when it was more likely in response to the reward itself. Another methodological difference between previous sign-tracking studies and the present study is the inclusion of a lever associated with the CS. Previous studies tested the animals in a Pavlovian conditioning paradigm after associating the CS with a lever (Fitzpatrick & Morrow, 2016; Meyer et al., 2012; Paolone et al., 2013; Stead et al., 2006). This such method may allow for a more accurate measure of sign responding than what was used in the present study, which measured sign responding as contacts with the light cue. Within this method, random paw contacts that are an artifact of exploration may have been counted as sign responses. Even though sign responding within the present study was minimal, it cannot be confidently concluded that the small number of sign responses that were observed were actually the attribution of incentive salience to the light cue or merely an artifact of animal movement.

Even if the paradigm used in the present study was optimal, and the results did demonstrate the emergence of sign- and goal- tracking phenotypes, it is important to consider the potential limitations of measuring sign-tracking behaviour as a dependent variable of a maternal immune activation manipulation. Though the literature

highlights common neurobiological mechanisms implicated in sign-tracking and the disorders commonly modelled by maternal infection, there are several assumptions that should be confirmed before exploring this relationship. Firstly, it may be beneficial to investigate the potential effect of maternal immune activation on simpler operant and Pavlovian processes before exploring the effect on response phenotype. In addition to no significant effect being found in the present study, it cannot be concluded that the light cue was successfully paired with the reward delivery and Pavlovian conditioning occurred at all. The method did not include an assessment of the strength of the association between the cue and reward. This would be a worthy inclusion in future studies investigating this relationship as it would allow for group differences in the acquisition of pavlovian conditioning to be detected. Studies investigating reward learning in ASD (Dichter et al., 2012; Kohls et al., 2013; Scott-Van Zeeland et al., 2010) demonstrated that ASD populations characteristically show impairments in acquiring stimulus-reward associations. This could mean that the impairments that the present study aimed to investigate may have caused group differences in the learning procedure alone, such that there was no conditioned response to be measured by the procedure. It is unknown whether this occurred in the present study as the method did not allow for such differences to be detected. As well as methodological changes, further research is required to establish the paradigm specifics, such as number of trials, necessary to achieve Pavlovian conditioning.

Another important factor to consider when researching reward learning in neurodevelopmental populations, such as ASD and schizophrenia, is stimulus type. There are obvious limitations to the extent this can be adjusted in animal studies, but it is still important to investigate the differences in reward learning across the available stimulus types (food, drug, social) in maternal infection populations. For example, the

literature would suggest that drug cues would be highly reinforcing in schizophrenia populations than a less-rewarding alternative (e.g. food) (Chambers et al., 2001; Kalivas & Volkow, 2005; Menne & Chesworth 2020). This would suggest that such a discrepancy in reward value would influence the acquisition of stimulus-reward pairings. Similarly, much of the ASD reward learning literature (Dichter et al., 2012; Kohls et al., 2013; Scott-Van Zeeland et al., 2010) focuses on the different activation of reward circuits elicited by different stimulus types. Only once these nuances around the acquisition of conditioned responses are well understood in these populations could research start looking at sign/goal differences using the maternal infection model.

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and enrichment in spontaneous anxiety-related behaviors. *Behavior genetics*, 36(5), 697- 712.

Appendix A

ANOVA tables

ANOVA – p(sign)

	Sum of Squares	df	Mean Square	F	p
Treatment	0.00439	1	0.00439	0.3356	0.567
Sex	0.00118	1	0.00118	0.0903	0.766
Treatment*Sex	0.01170	1	0.1170	0.8939	0.353
Residuals	0.36665	28	0.01309		

ANOVA – total responses

	Sum of Squares	df	Mean Square	F	p
Treatment	2048	1	2048	0.376	0.545
Sex	26796	1	26796	4.919	0.035
Treatment*Sex	8581	1	8581	1.575	0.220
Residuals	152522	28	5447		